

MATTHEW SCANLAND

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PROFESSIONAL SUMMARY

Backend software engineer with 7+ years building production APIs and modernizing legacy systems. Hands-on Python, FastAPI, .NET, Azure, and SQL experience with technical leadership, cross-team delivery, and mentoring.

EXPERIENCE

SONIC AUTOMOTIVE | REMOTE

AUG 2023 - PRESENT

Software Engineer, Feb 2025 - Present | **Junior Software Engineer**, Aug 2023 - Feb 2025

- Led the design and implementation of DAP Data Ingestion, migrating prioritized rule sets from Rules Engine to eventually consistent Azure Storage Tables; reduced rule retrieval and application time from 4 seconds to 300 ms for approximately 10,000 contractual offers per day.
- Assumed technical leadership of DAP during a team transition and delivered planned work on schedule; now co-lead DAP, lead the Rules Engine, and coordinate pricing-logic migration with Data Analytics and Pricing Engineering.
- Build Python/FastAPI services and REST APIs for internal teams and vehicle-purchasing partners; deploy on Azure App Service behind API Management with Cosmos DB and co-design Terraform for Azure Functions.
- Modernize the legacy .NET, SQL Server, and IIS Rules Engine through Self Service; built the initial Vue interface to improve users' ability to view and create rules, and integrate Azure Functions to improve maintainability, search, and rule processing.
- Own production releases and incident response for DAP, Rules Engine, and Self Service.
- Mentor and guide 4 engineers; one became the primary developer for a new valuation service.
- Create detailed engineering tickets that keep refinement meetings focused and let engineers weigh in on implementation.

VIRGINIA TECH - CENTER FOR POWER ELECTRONICS SYSTEMS (CPES)

APR 2019 - SEP 2023

Junior Software Engineer, Mar 2021 - Sep 2023 | **Full Stack Engineer**, Apr 2019 - Mar 2021

- Architected and maintained a PHP and MariaDB internal platform as CPES's sole software developer, serving hundreds of daily users; migrated HTML and Google Apps Script workflows to a custom MVC framework and REST API with multi-factor authentication and role-based access.
- Built the Lab Validation System across planning, schema design, implementation, and maintenance; integrated billing, leave, signatures, exports, and role-specific workflows, cutting validation time to 25% of the previous process.
- Delivered purchasing and reimbursement applications processing 10,000+ requests annually, with SQL-backed approval workflows, search, and near-real-time reporting.
- Designed an Azure cold-storage data lake roadmap for 35 TB and approximately 15 million files, evaluating ingestion, encryption, access control, migration, redundancy, and cost.
- Deployed and maintained Docker containers, Apache web servers, and Linux and Windows infrastructure; automated LAMP file structures and database backups with Bash.

TECHNICAL SKILLS

PROGRAMMING: Python, C#, SQL, PHP, JavaScript, Bash

BACKEND: FastAPI, .NET, REST APIs, API design, object-oriented programming, legacy modernization

CLOUD AND INFRASTRUCTURE: Azure, AWS, Kubernetes, Docker, Terraform, Azure Storage, Linux, Git, CI/CD

DATA AND WEB: Azure Cosmos DB, SQL Server, MySQL/MariaDB, Vue, Angular, IIS, Apache

EDUCATION

VIRGINIA TECH

Bachelor of Science in Computer Science